

확률·랜덤프로세스 기말대비 선별 5문항 — Lecture 09 표본평균·WLLN·CLT(정규근사·연속성보정) (★=핵심)

① 표본평균이면 $E=\mu, \text{Var}=\sigma^2/n$ ② 확률 물으면 CLT 표준화→ Φ (Q함수), 이산은 ± 0.5 보정 ③ 채점: origin 풀이 / /solve-problem

떠올릴 개념

- 표본평균 $M_n=(X_1+\dots+X_n)/n$: $E[M_n]=\mu$, $\text{Var}(M_n)=\sigma^2/n$ ($n \uparrow$ 분산 $\downarrow$)
- WLLN(약한큰수): $M_n \rightarrow \mu$ (확률수렴, in prob.) — 임의 $\epsilon>0$ 에 $P(|M_n-\mu|>\epsilon) \rightarrow 0$
- CLT: $(\sum X_i - n\mu)/(\sigma\sqrt{n}) \rightarrow N(0,1)$ (분포수렴) — 표본합·표본평균을 정규로 근사
- 확률 계산: 표준화 후 Φ 표(또는 $Q=1-\Phi$)로 $P(\dots)$ 환산, $\sigma_{\{M_n\}}=\sigma/\sqrt{n}$
- 이산RV(예: 이항) 정규근사 시 연속성 보정 ± 0.5 (예 $P(K \leq k) \approx \Phi((k+0.5-\mu)/\sigma)$)
- 이 과목 미출제: Chebyshev-MGF·강한LLN-a.s./L 수렴 (정규근사/표본평균만 사용)
- 모든 PMF/PDF는 명시 구간 밖에서 otherwise = 0

문항 목차 (총 5문항)

- [1] L09 · PSP 10.1.1 — [●] 표본평균 분산 $\text{Var}[M_9]=\sigma^2/9 \cdot P[X_1>7]$ · CLT $P[M_9>7]$
- [2] L09 · IPSRP 7.3.7 — [] 측정 정밀도용 표본크기 n 결정 $P(|M_n-q|\leq 0.1)\geq 0.95$ — CLT 정규근사
- [3] L09 · IPSRP 7.3.5 — [] CLT 표본합 정규근사: 7시간에 ≤ 40 job 처리확률 ★
- [4] L09 · PSP 9.5.2 — [●] CLT 이산(이항)근사·연속성보정: $P[16 \leq K_{100} \leq 24]$ ★
- [5] L09 · IPSRP 7.3.4 — [] CLT 연속성보정 명시: 50명 중 주차>30대 확률

10.1.1 ● $X_1, \dots, X_n$ is an iid sequence of exponential random variables, each with expected value 5.

- (a) What is $\text{Var}[M_9(X)]$, the variance of the sample mean based on nine trials?
- (b) What is $P[X_1 > 7]$, the probability that one outcome exceeds 7?
- (c) Use the central limit theorem to estimate $P[M_9(X) > 7]$, the probability that the sample mean of nine trials exceeds 7.

Problem 7

An engineer is measuring a quantity q . It is assumed that there is a random error in each measurement, so the engineer will take n measurements and reports the average of the measurements as the estimated value of q . Specifically, if Y_i is the value that is obtained in the i 'th measurement, we assume that

$$Y_i = q + X_i,$$

where X_i is the error in the i th measurement. We assume that X_i 's are i.i.d. with $EX_i = 0$ and $\text{Var}(X_i) = 4$ units. The engineer reports the average of measurements

$$M_n = \frac{Y_1 + Y_2 + \dots + Y_n}{n}.$$

How many measurements does the engineer need to make until he is 95% sure that the final error is less than 0.1 units? In other words, what should the value of n be such that

$$P(q - 0.1 \leq M_n \leq q + 0.1) \geq 0.95 \quad ?$$

— 풀이 —

Problem 5

The amount of time needed for a certain machine to process a job is a random variable with mean $EX_i = 10$ minutes and $\text{Var}(X_i) = 2$ minutes². The times needed for different jobs are independent from each other. Find the probability that the machine processes less than or equal to 40 jobs in 7 hours.

9.5.2● Internet packets can be classified as video (V) or as generic data (D). Based on a lot of observations taken by the Internet service provider, we have the following probability model: $P[V] = 3/4$, $P[D] = 1/4$. Data packets and video packets occur independently of one another. The random variable K_n is the number of video packets in a collection of n packets.

- (a) What is $E[K_{100}]$, the expected number of video packets in a set of 100 packets?
- (b) What is $\sigma_{K_{100}}$?
- (c) Use the central limit theorem to estimate $P[K_{100} \geq 18]$.
- (d) Use the central limit theorem to estimate $P[16 \leq K_{100} \leq 24]$.

Problem 4

50 students live in a dormitory. The parking lot has the capacity for 30 cars. Each student has a car with probability $\frac{1}{2}$, independently from other students. Use the CLT (with continuity correction) to find the probability that there won't be enough parking spaces for all the cars.
